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AI-Based QR Code Fraud and Scam Detection System for Secure Digital Payments

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ABSTRACT - The rapid proliferation of QR code-based digital payment systems in India, accelerated by the Unified Payments Interface (UPI) ecosystem, has been accompanied by a sharp rise in QR code-based fraud schemes. The Reserve Bank of India's Annual Report on Digital Payment Frauds documented that UPI-related fraud cases increased by 85% between fiscal years 2022 and 2024, with QR code manipulation, fake merchant codes, and quishing (QR code phishing) attacks emerging as dominant attack vectors, particularly targeting small merchants, elderly users, and first-time digital payment adopters with limited technical literacy. Conventional QR code scanning applications perform no security validation beyond basic format parsing, leaving users exposed to malicious payloads embedded within seemingly legitimate-looking codes. This paper presents an AI-Based QR Code Fraud and Scam Detection System that performs multi-layered real-time security analysis at the point of QR code scanning, combining a Convolutional Neural Network-based visual tamper detection module that identifies physically altered or overlaid QR codes, a URL and payment payload risk classifier using a fine-tuned BERT-based natural language model trained on a labelled corpus of fraudulent and legitimate UPI payment strings, a merchant verification layer that cross-references VPA (Virtual Payment Address) and merchant identifiers against the National Payments Corporation of India's verified merchant registry, and a behavioral anomaly detection module that flags unusual payment patterns such as sudden high-value requests from previously low-value merchant codes. Evaluated against a dataset of 38,400 QR codes comprising verified legitimate merchant codes and documented fraud samples collected in collaboration with cybercrime cell reports from three Tamil Nadu districts, the proposed system achieves a fraud detection accuracy of 96.7%, false positive rate of 2.3%, and end-to-end scan-to-verdict latency of 340 milliseconds, demonstrating strong viability for integration into existing UPI-enabled payment applications.

Keywords *— QR Code Security, UPI Fraud Detection, Quishing, BERT, CNN, Digital Payment Security, Phishing Detection, Merchant Verification, Behavioral Anomaly Detection, Fintech Security*

I. INTRODUCTION

India's digital payment ecosystem has undergone a transformation of global significance over the past decade, anchored by the Unified Payments Interface (UPI), which the National Payments Corporation of India reports processed over 131 billion transactions valued at approximately Rs. 200 lakh crore during the2023–24 fiscal year. Quick Response (QR) codes have emerged as the dominant interface mechanism for UPI-based merchant payments, owing to their low infrastructure cost, ease of generation, and rapid scan-to-pay user experience that has driven adoption across street vendors, small retail establishments, and informal sector merchants who previously lacked access to digital payment acceptance infrastructure.

This success, however, has created a correspondingly large attack surface for financially motivated fraud. The Reserve Bank of India's Annual Report on Digital Payment Frauds for FY 2023–24 documented that UPI-related fraud complaints rose 85% over the previous two fiscal years, with QR code-based attacks constituting a rapidly growing fraud category. These attacks manifest in several distinct forms: physically tampered QR codes where a fraudulent sticker is overlaid on a legitimate merchant's printed QR code at the point of sale, redirecting payments to an attacker-controlled account; quishing attacks where malicious QR codes embedded in emails, posters, or social media direct victims to phishing pages that harvest UPI credentials; fake "refund" or "cashback" QR codes that, when scanned, initiate a payment request rather than a receipt (exploiting the visual similarity between UPI collect requests and payment requests for unsophisticated users); and counterfeit merchant QR codes that mimic the visual branding of legitimate payment aggregators while routing funds to fraudulent accounts.

These attack vectors disproportionately affect precisely the user populations that UPI's accessibility-driven design was intended to serve: small and informal sector merchants with limited cybersecurity awareness, elderly users who are frequent targets of social engineering, and first-time digital payment adopters in rural and semi-urban areas who lack the

technical literacy to scrutinize a QR code's underlying payload before scanning. Existing QR code scanning applications, including the camera-integrated scanners bundled with most UPI payment apps, perform essentially no security validation they decode the QR code's embedded data and present the resulting payment request to the user without any assessment of the legitimacy of the destination VPA, the consistency of the transaction request with the merchant's typical transaction profile, or visual indicators that the code itself may have been physically tampered with.

This paper presents an AI-Based QR Code Fraud and Scam Detection System designed to close this security gap by performing comprehensive, multi-layered risk analysis at the moment of QR code scanning before the user is presented with a payment confirmation screen. The system integrates four complementary detection layers, each targeting a distinct fraud vector, and is designed for integration as a software development kit (SDK) into existing UPI-enabled payment applications without requiring changes to NPCI's underlying UPI protocol infrastructure.

The key contributions of this paper are:

- Design of a four-layer AI-based QR code fraud detection architecture combining visual tamper detection, payload risk classification, merchant identity verification, and behavioral anomaly detection.
- A CNN-based visual tamper detection model trained to identify physical alterations, overlay stickers, and printing inconsistencies in photographed QR codes, distinguishing genuine merchant-printed codes from fraudulently modified ones.
- A fine-tuned BERT-based UPI payload risk classifier trained on a labelled corpus of 38,400 legitimate and fraudulent QR payment strings, capable of identifying suspicious VPA patterns, malformed payment parameters, and known fraud-associated payload structures.
- Integration of real-time merchant verification against NPCI's verified merchant registry and a behavioral anomaly detection layer that flags transaction requests inconsistent with a merchant's historical payment profile.
- Comprehensive evaluation using a real-world dataset assembled in collaboration with cybercrime cell fraud reports from three Tamil Nadu districts, demonstrating 96.7% detection accuracy at sub-400-millisecond latency suitable for real-time payment app integration.

II. LITERATURE REVIEW

Research on QR code security spans three overlapping domains: general phishing and malicious URL detection, QR code-specific attack taxonomies, and machine learning approaches to payment fraud detection. Vidas et al. [1] conducted one of the earliest systematic studies of QR code-based attack vectors, documenting attack patterns including URL redirection to phishing sites and malicious app installation triggers, and established a foundational taxonomy of QR code attack types that informed subsequent security research in this domain.

Krombholz et al. [2] conducted a large-scale empirical study of QR code usage and security awareness, finding that the overwhelming majority of users scan QR codes without any verification of the destination URL or payload, and that visual design cues intended to convey legitimacy (such as branded QR code frames) had limited correlation with actual code legitimacy —a finding that directly motivates the present work's emphasis on automated, rather than user-judgment-dependent, security validation.

Machine learning-based phishing URL detection has been extensively studied in the broader cybersecurity literature. Sahingoz et al. [3] applied multiple machine learning classifiers including Random Forest, Naive Bayes, and Artificial Neural Networks to phishing URL detection using natural language processing-derived features extracted from URL strings, demonstrating that lexical and structural URL features (subdomain count, special character frequency, domain age proxies) provided strong discriminative signal without requiring access to the destination page content. Le et al. [4] proposed URLNet, a deep learning architecture combining character-level and word-level CNN representations of URLs, achieving improved detection accuracy over feature-engineered approaches by learning representations directly from raw URL text.

Transformer-based approaches to text classification, including phishing and fraud detection, gained prominence following the introduction of BERT by Devlin et al. [5]. Maneriker et al. [6] adapted a BERT-style architecture specifically for URL classification (URLTran), demonstrating that pretraining on large URL corpora followed by fine-tuning on labelled phishing datasets substantially outperformed both classical machine learning and earlier deep learning baselines for phishing URL detection, establishing the architectural pattern adopted for the present work's payload risk classifier.

In the specific domain of QR code and mobile payment fraud, Focardi et al. [7] analyzed security vulnerabilities in mobile payment QR code implementations and identified the absence of cryptographic integrity verification in most deployed QR

payment systems as a structural vulnerability enabling tampering attacks a finding consistent with the visual tamper detection requirement addressed by the present work's CNN-based module. Bahnsen et al. [8] applied deep learning to URL-based phishing detection with a specific focus on detecting visually similar (typosquatting) domain names designed to impersonate legitimate financial institutions, a technique directly applicable to detecting fraudulent merchant identifiers designed to resemble legitimate UPI VPAs.

Behavioral and transaction pattern-based fraud detection has been studied extensively in the broader payment fraud literature. Dal Pozzolo et al. [9] applied undersampling and cost-sensitive learning techniques to credit card fraud detection, addressing the severe class imbalance characteristic of fraud detection problems where fraudulent transactions represent a small fraction of total transaction volume a methodological consideration directly relevant to the present work's behavioral anomaly module, given that fraudulent QR scan attempts represent a small minority of total scan volume in any deployed system.

III. PROBLEM STATEMENT

The current QR code-based UPI payment ecosystem exhibits the following critical security gaps that directly enable the fraud patterns documented in RBI's annual fraud reports:

- Absence of Visual Tamper Detection: No widely deployed UPI payment application verifies whether a scanned QR code shows signs of physical tampering, such as an overlay sticker placed atop a legitimate merchant's printed code — a common attack vector against street vendors and small retail merchants.
- No Payload-Level Risk Assessment: QR code scanning applications decode and act on the embedded payment payload without assessing whether the destination VPA, payment amount field, or transaction note exhibits patterns associated with known fraud schemes.
- Lack of Real-Time Merchant Identity Verification: There is no readily accessible, real-time mechanism for an end-user's payment app to verify that the VPA embedded in a scanned QR code corresponds to a registered, verified merchant rather than an individual or fraudulent account.
- No Behavioral Context for Transaction Requests: Payment requests are processed without reference to the requesting merchant's historical transaction profile, meaning a QR code that has processed only small daily transactions can suddenly request an unusually large payment without triggering any review.
- Disproportionate Impact on Vulnerable Users: The combined effect of these gaps falls hardest on users least equipped to manually detect fraud indicators — elderly individuals, first-time digital payment users, and small merchants without dedicated IT or security staff.

The proposed system addresses each of these gaps through an integrated, automated, real-time detection architecture detailed in the following section.

IV. PROPOSED SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

The proposed system operates as a four-layer detection pipeline integrated at the point of QR code scanning within a UPI payment application, processing each scanned code through visual analysis, payload analysis, merchant verification, and behavioral analysis before presenting a risk-annotated payment confirmation to the user.

B. Layer 1: CNN Visual Tamper Detection

When a user's camera captures a QR code image, the raw image frame prior to QR decoding is passed through a lightweight MobileNetV3-Small CNN trained to detect visual indicators of physical tampering. The model is trained on a dataset of 14,200 QR code images comprising genuine, undamaged merchant-printed codes and a constructed set of tampered samples generated by digitally compositing overlay stickers, printed labels, and adhesive patches at varying positions, rotations, and lighting conditions onto genuine QR code base images, supplemented with 380 real-world tampered QR code photographs collected from documented fraud cases shared by Tamil Nadu Police Cyber Crime Wing. The model detects tampering indicators including edge discontinuities at sticker boundaries, inconsistent print resolution or color profile between regions of the code, shadow patterns inconsistent with a single flat printed surface, and partial occlusion patterns characteristic of overlay stickers that do not perfectly align with the original code's module grid.

C. Layer 2: BERT-Based Payload Risk Classification

Once the QR code is successfully decoded, the resulting UPI payment string (in the standard upi://pay?pa=...&pn=...&am=...&tn=... format) is tokenized and passed through a fine-tuned BERT-base model adapted for structured payment string classification rather than natural language text. The model treats the UPI string's component fields payee VPA, payee name, amount, and transaction note as a structured sequence and is fine-tuned on a labelled corpus of 38,400 payment strings, of which 31,600 are verified

legitimate merchant codes collected through partnership with regional payment aggregators and 6,800 are documented fraudulent payment strings sourced from cybercrime cell case records across three Tamil Nadu districts (Tiruvannamalai, Vellore, and Cuddalore). The classifier learns to identify suspicious patterns including VPA handles using look-alike characters or domains mimicking known payment aggregators, payee names inconsistent with the claimed merchant category, amount fields containing unusual precision or values inconsistent with typical small-merchant transaction ranges, and transaction note fields containing urgency-inducing or social-engineering language patterns ("refund", "prize", "verify account") commonly associated with quishing schemes.

D. Layer 3: NPCI Merchant Registry Verification

For payment requests directed to a VPA registered as a merchant account (as opposed to a peer-to-peer personal VPA), the system performs a real-time lookup against the NPCI Merchant Verification API, cross-referencing the decoded VPA against the registry of verified merchants holding valid merchant category codes (MCC) and KYC-compliant business registration. A mismatch between the QR code's claimed merchant name (embedded in the payload's pn field) and the registry's verified business name, or a VPA pattern matching characteristics of accounts flagged in NPCI's shared fraud intelligence database, elevates the transaction's risk score substantially. For peer-to-peer VPAs without merchant registration, this layer instead checks the VPA against a continuously updated shared fraud reporting database maintained collaboratively across participating banks and payment service providers under the National Cyber Crime Reporting Portal's data-sharing framework.

E. Layer 4: Behavioral Anomaly Detection

For merchant VPAs with sufficient transaction history available to the scanning application (either through the user's own prior transaction history with that merchant or through aggregated, privacy-preserving transaction pattern statistics shared across the payment ecosystem), an Isolation Forest anomaly detection model flags transaction requests that deviate substantially from the merchant's established pattern. Key behavioral features include the requested amount relative to the merchant's historical transaction amount distribution, the time-of-day and day-of-week consistency with the merchant's typical operating pattern, and the velocity of QR code regeneration (a merchant whose displayed QR code has changed unusually frequently in a short period may indicate code tampering or account compromise). This layer is calibrated to minimize false positives for legitimate business pattern changes (such as a restaurant's higher average transaction value during a festival season) by incorporating a rolling 90-day baseline that adapts to gradual legitimate pattern shifts while still flagging abrupt anomalies.

F. Composite Risk Scoring and User Presentation

The outputs of all four layers are combined into a composite risk score using a weighted ensemble: 25% visual tamper score, 35% payload risk classification score, 25% merchant verification score, and 15% behavioral anomaly score, with weights determined through validation set optimization to minimize overall false negative rate while maintaining false positive rate below 3%. The composite score maps to one of three user-facing verdicts: SAFE (proceed with standard payment flow), CAUTION (display a warning banner with the specific risk indicators detected, requiring explicit user acknowledgment before proceeding), or BLOCK (prevent the transaction from proceeding and direct the user to report the code, reserved for composite scores indicating high-confidence fraud).

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Dataset and Evaluation Setup

The system was evaluated on a dataset of 38,400 QR codes comprising 31,600 verified legitimate merchant payment codes collected through partnership with regional payment aggregators operating in Tamil Nadu, and 6,800 documented fraudulent QR codes and payment payloads sourced from cybercrime cell case records across Tiruvannamalai, Vellore, and Cuddalore districts, covering the period 2021–2024. An 80/10/10 split was applied for training, validation, and testing, with careful stratification to ensure fraud sub-types (tampered codes, quishing payloads, fake refund requests, counterfeit merchant codes) were proportionally represented across all splits.

B. Overall Detection Performance

TABLE I. Fraud Detection Performance - Full System vs. Individual Layers

Detection Configuration	Accuracy (%)	Recall (%)	False Positive (%)	Latency (ms)
Layer 1 only (Visual Tamper)	78.4	71.2	4.8	95
Layer 2 only (BERT Payload)	89.6	87.3	5.1	140
Layer 3 only (Merchant Verify)	82.1	79.8	3.4	180

Layer 4 only (Behavioral)	74.9	69.5	6.2	60

The full integrated system achieves 96.7% detection accuracy and 95.4% recall, substantially outperforming any individual detection layer in isolation. The BERT-based payload classifier (Layer 2) is the strongest individual contributor at 89.6% accuracy, reflecting that the majority of documented fraud cases in the dataset involved identifiable payload-level indicators such as suspicious VPA patterns or social-engineering transaction notes. However, the combination of all four layers reduces the false positive rate from a range of 3.4–6.2% for individual layers down to 2.3% for the integrated system, demonstrating that the layers provide complementary rather than redundant signal a tampered code without payload-level red flags, for instance, is still caught through Layer 1, while a clean-looking code with an unregistered merchant VPA is caught through Layer 3.

C. Performance by Fraud Sub-Type

TABLE II. Detection Recall by Fraud Sub-Type

Fraud Sub-Type	Recall (%)	Sample Count	Primary Detecting Layer
Physically tampered (overlay sticker)	93.8	1,240	Layer 1 (Visual)
Quishing (phishing redirect)	97.2	2,180	Layer 2 (Payload)
Fake refund/cashback request	96.5	1,640	Layer 2 (Payload)
Counterfeit merchant code	94.1	980	Layer 3 (Verify)
Compromised legitimate merchant	89.7	760	Layer 4 (Behavioral)
Overall Weighted Average	95.4	6,800	All

Quashing attacks show the highest recall at 97.2%, attributable to the strong lexical and structural signal present in malicious VPA and transaction note fields that the BERT classifier readily learns to identify. Compromised legitimate merchant accounts where a previously genuine merchant's

Full System (All 4 Layers)	96.7	95.4	2.3	340

UPI credentials have been hijacked and used to request fraudulent payments show the lowest recall at 89.7%, reflecting the inherent difficulty of this fraud sub-type: because the underlying VPA and merchant registration are legitimate, only Layer 4's behavioral anomaly detection can flag the abrupt deviation from established transaction patterns, and this signal is necessarily weaker than the more direct visual or payload-level indicators available for other fraud types.

D. Latency Breakdown and Real-Time Viability

TABLE III. Per-Layer Processing Latency Breakdown

Processing Stage	Mean Latency (ms)	95th Percentile (ms)	Device
QR decode + image crop	25	40	On-device
Layer 1 — CNN tamper check	70	95	On-device
Layer 2 — BERT payload check	115	160	Cloud API
Layer 3 — NPCI registry lookup	95	180	Cloud API
Layer 4 — Behavioral scoring	35	50	Cloud API
Total End-to-End (parallelized)	340	420	Hybrid

The end-to-end mean latency of 340 milliseconds is achieved through parallelized execution of Layers 1-4 following the initial QR decode step, rather than sequential processing Layer 1 runs on-device immediately upon image capture while Layers 2–4 are dispatched to cloud APIs concurrently, with the composite risk score computed once all four results are received. This latency profile is well within the threshold for seamless integration into existing payment app scan-to-confirmation flows, where a delay of under 500 milliseconds

is generally imperceptible to users as a distinct processing step rather than as a degradation of app responsiveness.

VI. CONCLUSION

This paper presented an AI-Based QR Code Fraud and Scam Detection System that addresses the rapidly growing problem of QR code-based UPI payment fraud through an integrated, four-layer detection architecture combining visual tamper detection, payload risk classification, merchant registry verification, and behavioral anomaly detection. The system is specifically designed to protect the user populations most vulnerable to QR code fraud small merchants, elderly users, and first-time digital payment adopters who are least equipped to manually detect the visual and contextual indicators of fraudulent codes.

Evaluation on a real-world dataset assembled in collaboration with cybercrime cell records from three Tamil Nadu districts demonstrated 96.7% overall detection accuracy, 95.4% recall across diverse fraud sub-types, and a false positive rate of 2.3% low enough to avoid significant disruption to legitimate transaction flows while still providing meaningful fraud protection. The layer-wise ablation analysis confirmed that each of the four detection layers contributes complementary signal, with the combination substantially outperforming any individual layer, and the sub-400-millisecond end-to-end latency confirms the system's practical viability for integration into existing UPI payment applications without perceptible impact on user experience.

Future work will focus on expanding the fraud training dataset through continued collaboration with cybercrime cells across additional Indian states to improve generalization beyond the Tamil Nadu-specific fraud patterns reflected in the current dataset, exploring federated learning approaches that would allow the behavioral anomaly detection layer to benefit from cross-institutional transaction pattern data without centralizing sensitive financial information, and developing a lightweight on-device variant of the BERT payload classifier using knowledge distillation to reduce dependence on cloud connectivity for users in low-connectivity rural areas where quishing attacks targeting offline-then-online payment flows present a distinct emerging risk.

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